APS106

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introduction to Object Oriented Programming, and the file object.

Week 5 | Lecture 2 (5.2)

While waiting, download the Jupyter Notebook and text file for today's lecture

if nothing else, write #cleancode

This Week's Content

- **Lecture 5.1**
  - Strings: Conversions, Indexing, Slicing, and Immutability
- **Lecture 5.2**
  - **Strings: practice exercise**
  - **Introduction to Object-Oriented Programming and the File Object**
- **Lecture 5.3**
  - **for loops**

Procedural Programming

Global Variable
Pedestrian 1
x, y Location

Global Variable
Pedestrian 2
x, y Location

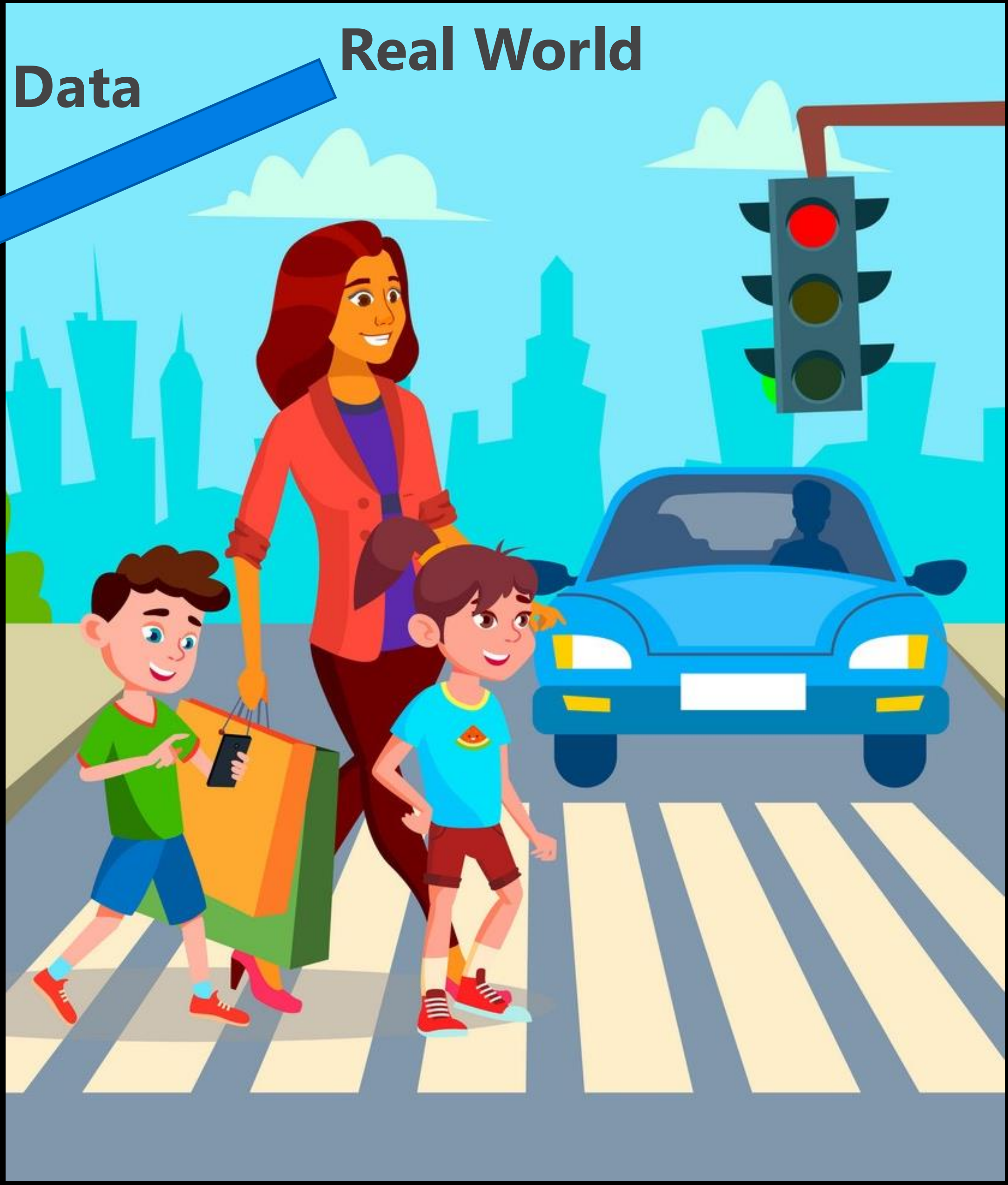
Global Variable
Pedestrian 3
x, y Location

```
x_ped1 = 3
y_ped1 = 5
```

Global Variable
Traffic Light 1
Color

Global Variable
Car 1
x, y location

```
traffic_light = 'red'
```



Separation of Data and Functions

Function
`okToCross`

Function
`AdvancePosition`

Function
`isInIntersection`

Procedural Programming

Global Variable
Pedestrian 1
x, y Location

Global Variable
Pedestrian 2
x, y Location

Global Variable
Pedestrian 3
x, y Location

Global Variable
Traffic Light 1
Color

Global Variable
Car 1
x, y location

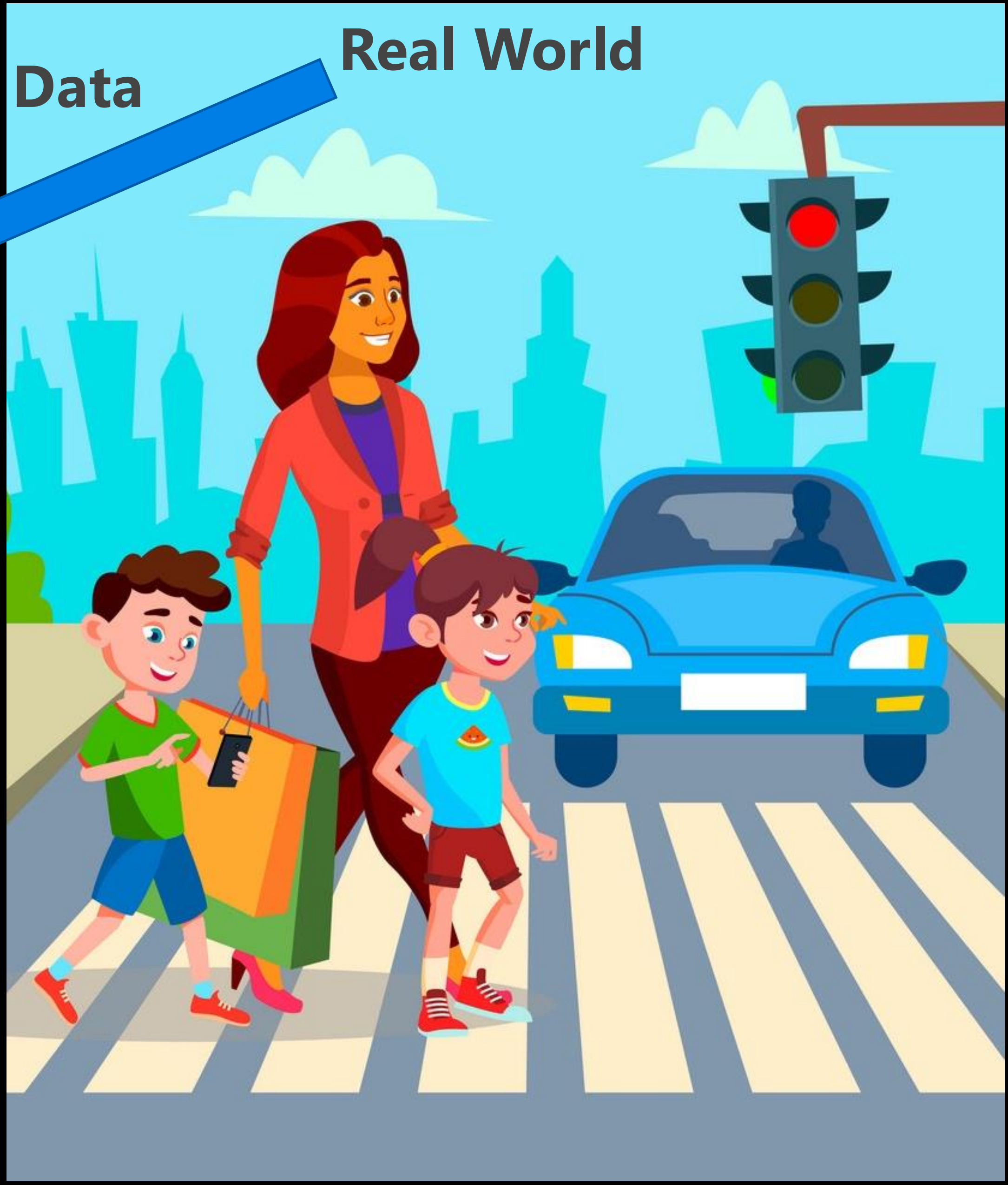
input

output

Function
`okToCross`

Function
`AdvancePosition`

Function
`isInIntersection`



Object-Oriented Programming

Object-oriented programming bundle the pedestrian's Data together with these Functions into a single object

Car Object
x, y Location
Pedestrian in Intersection
Advance Position

Traffic Light Object
Color
Change Color

Pedestrian Object
x, y Location
Ok to Cross
Advance Position

Object: Pedestrian

- **Data**: position, status
- Methods:
    - okToCross()
    - advancePosition()

Object-Oriented Programming

- Often, an object definition corresponds to some object or concept in the real world.
- The functions that operate on that object correspond to the ways real-world objects interact.

- Models our real-life thinking
  - Sandwich – ingredients, freshness, etc.
  - Car – model, year, fuel level, forward, reverse etc.
  - Cat – weight, name, colour, scratch, meow, sleep, etc.
  - Turtle Object – x_position, y_position, forward, up, left, etc.

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Object-Oriented Programming

Data

Functions

```
def up(y):
  return y + 1

def goto(x_new, y_new):
  return x_new, y_new

def right(x):
  return x + 1
```

Procedural

```
alex_x = 0
alex_y = 0

alex_y = up(alex_y)
alex_x, alex_y = goto(-150, 100)
alex_x = right(alex_x)

print(alex_x, alex_y)
```

Object-Oriented

```
alex = Turtle(0, 0)

alex.up()
alex.goto(-150, 100)
alex.right()

print(alex.x, alex.y)
```

Objects in Python

- **Everything in Python is an object.**
- Every value, variable, function, etc., is an object.
- Every time we create a variable we are making a new object.

Is this an instance of this class.

```
>>> isinstance(4, object)
True

>>> isinstance(max, object)
True

>>> isinstance("Hello", object)
True
```

Classes

- A template or blueprint for object types
- An instance of a class refers to one object whose type is defined as the class.
- The words "instance" and "object" are used interchangeably.
- A Class is made up of attributes (**data**) and methods (**functions**).

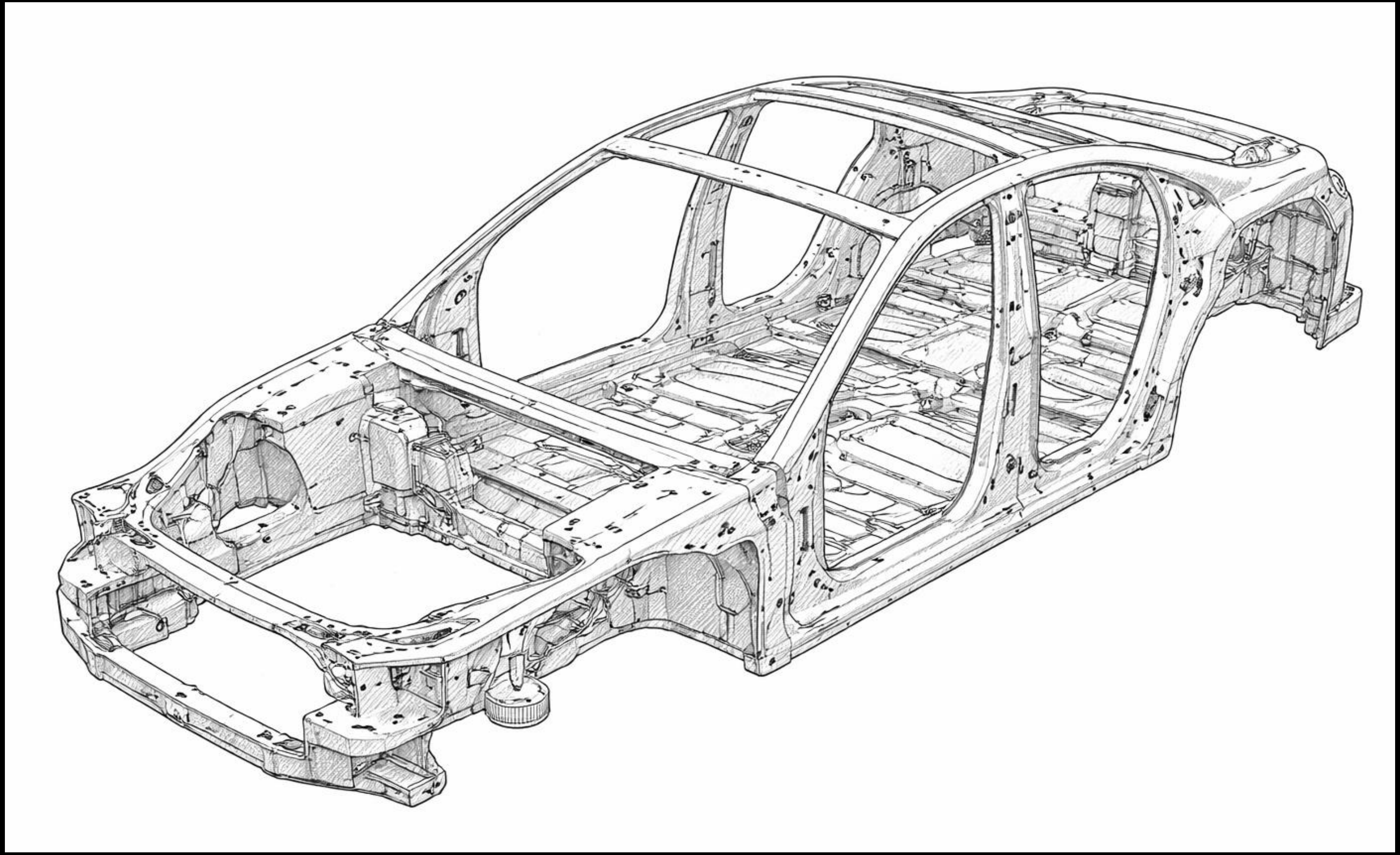
Class

Data → **Attributes**

Functions → **Methods**

```
str.upper('hello')
'hello'.upper()
```

Classes: blueprints

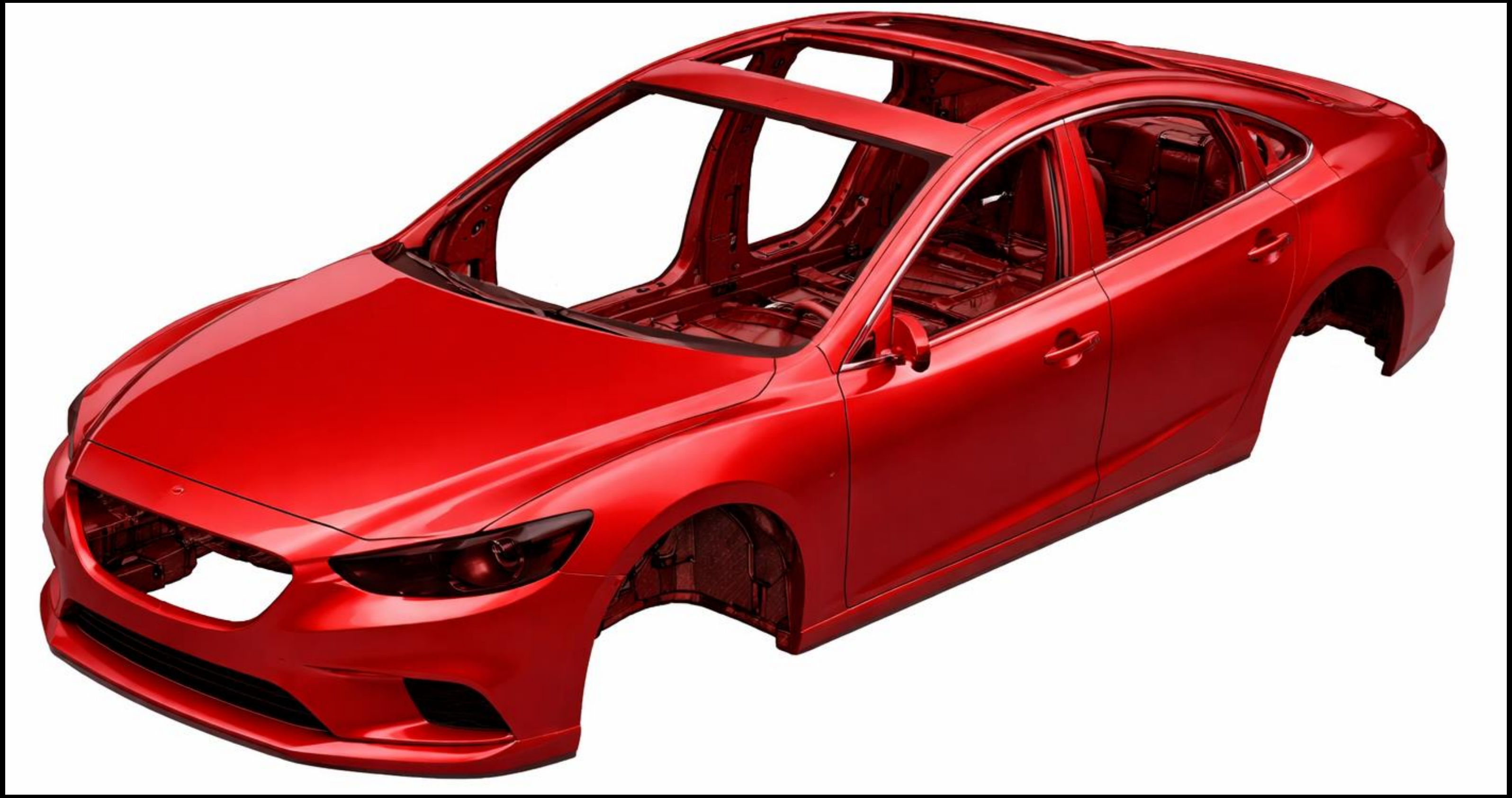


Car

model
company
year
color

brake
accelerate
open trunk

model: Corolla
year: 2024
tires: None
color: red



Car

model
company
year
color

brake
accelerate
open trunk

model: Corolla
year: 2024
tires: None
color: grey

Car

model
company
year
color

brake
accelerate
open trunk

model: Corolla
year: 2024
tires: fancy
color: grey

Car

model
company
year
color

brake
accelerate
open trunk

model: Corolla
year: 2024
tires: monster
color: grey

Car

model
company
year
color

brake
accelerate
open trunk

model: Sierra
year: 2020
tires: monster
color: grey flame decals

Car

model
company
year
color

brake
accelerate
open trunk

We've already been using objects!

integer

is_integer
as_integer_ratio
to_bytes
from_bytes
bit_count
...

string

replace
find
rfind
upper
isupper
islower
capitalize
count
...

math

pi
e
inf
...

factorial
ceil
floor
isfinite
log
...

Classes

We don't need to know these details to be able to use them!

We will learn more about building custom classes later…

```
alex = Turtle(0, 0)



alex.up()
alex.goto(-150, 100)
alex.down()

print(alex.x, alex.y)
```

```
class Turtle:

    def __init__(self, x, y):
        self.x = x
        self.y = y

    def up(self):
        body

    def goto(self, x, y):
        body

    def down(self):
        body
```

Classes

Instances (objects) of the **file** class.

name: itinerary.txt
mode: write

name: bills.txt
mode: append

name: grades.txt
mode: read

File

filename
mode (ex: read or write)

read
readline
write
close
...

Let’s Code!

- Let’s take a look at how this works in Python!
  - Objects in Python
  - One potential ACORN design

Breakout Session!

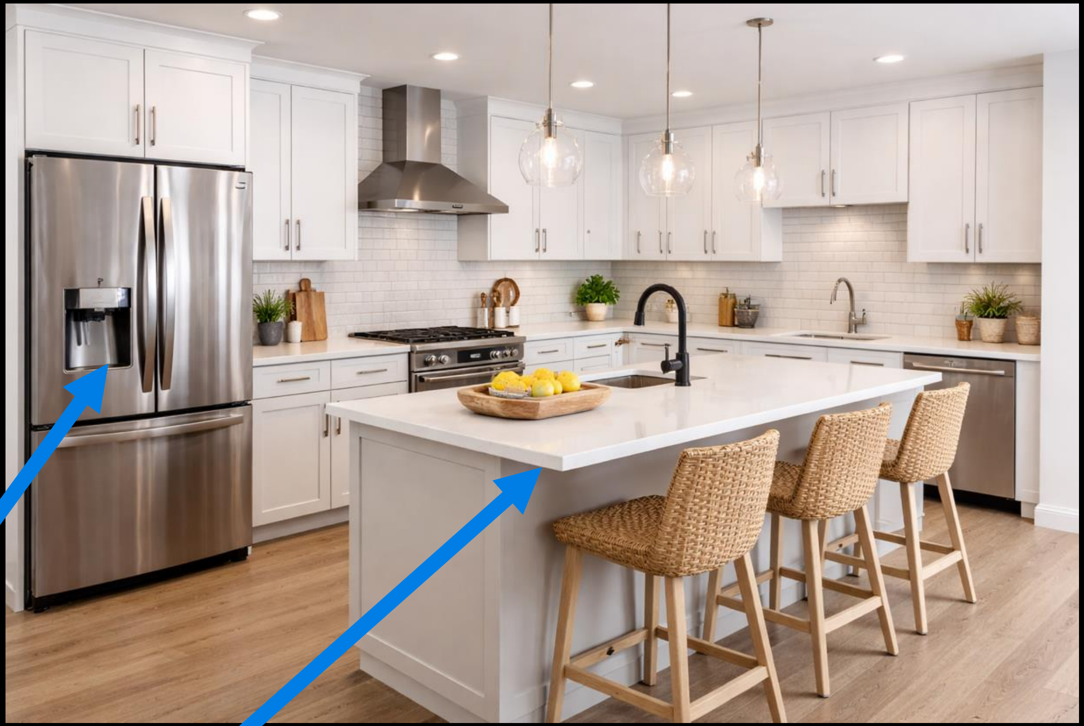
Click Link:
1. Objects in Python

Why do we care about files?

- Everything we've done so far is essentially deleted when our program ends
- What if we wanted to store information?

Why work with files?

- While a program is running, its data is stored in random access memory (RAM). *when the program ends data in RAM disappears*.
- To make data available the next time the program is started, it must be written to a non-volatile storage medium, such a hard drive, USB drive, cloud storage, or CD-R.



Disk

RAM

Why work with files?

- Data on non-volatile storage media are stored in named locations on the media called **files**.
- By **reading** and **writing** files, programs can save information between program runs.

It’s like working with a notebook!

- A file must be **opened**.
- When you are done, it has to be **closed**.
- While the file is open, it can either be **read from or written to**.
- Like a bookmark, the file **keeps track of** where you are reading to or writing from.
- You can read the whole file in its **natural order** or you can **skip** around

Opening a File

- Python has a built-in function `open` that creates and returns a file object with a connection between the file information on the disk and the program.
- The general form for opening a file is:

```
open(filename, mode)
```

- where **mode** is `'r'` (to open for reading), `'w'` (to open for writing), or `'a'` (to open for appending to what is already in the file).
- **filename** is an external storage location.

Writing to a File

- The following statement opens the file `test.txt` in write mode `‘w’`.

```
myfile = open(‘test.txt’, ‘w’)
```

- If there is no file named "`test.txt`" on the disk, it will be created. *<u>If there already is one, it will be replaced by the file we are writing</u>*.
- The file information will be assigned to the file object `myfile`.

Writing to a File

- To write something we need to use the `write` method as shown:

```
myfile.write('CATS!…')
```

- The `write` method does not attach a new line character by default.

```
myfile.write('\n')
myfile.write('I <3 my second line… \n')
```

- Every time we use `myfile.write()` a string is added to file "`test.txt`" where we left off.

```
myfile = open(‘test.txt’, ‘w’)
myfile.write(‘CATS!…’)
myfile.write(‘\n’)
myfile.write(‘I <3 my second sentence… \n’)

# and eventually we close our file
myfile.close()
```

Closing a File

- Once you have finished with the file, you need to close it:

```
myfile.close()
```

- This tells the system that we are done writing and makes the disk file available for reading or writing by other programs (or by our own program).

Let’s Code!

- Let’s take a look at how this works in Python!
  - Writing to your first file
  - Figure out where your files are getting saved!
    - Current working directory
  - Try playing with “w” and “a” mode!

Breakout Session!

Click Link:
2. The File Object

CLOSE THE FILE!

```
myfile.close()
```

- Always close a file that you've opened!
  - Many changes don't occur until **after** the file is closed
  - Leaving open is a waste of a computer's resources
    - Slows down program, uses more space in RAM, impacts performance
  - Other files may treat the file as "open" and won't be able to read the file
  - Could run into limits about how many files you have open
  - Not #cleancode

The "Writing to Files" Recipe

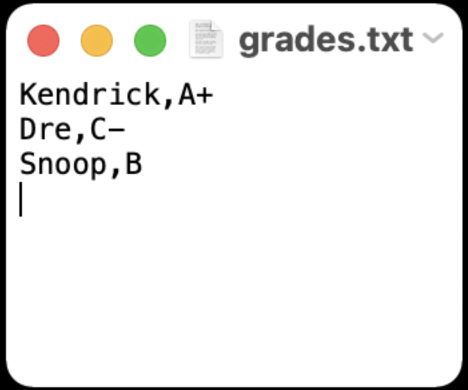
go together

```
# open/create a file
myfile = open("grades.txt", "w")

# write to a file
myfile.write('string')

# close the file
myfile.close()
```

Example: Writing to Files

- How would we write a string to a file?

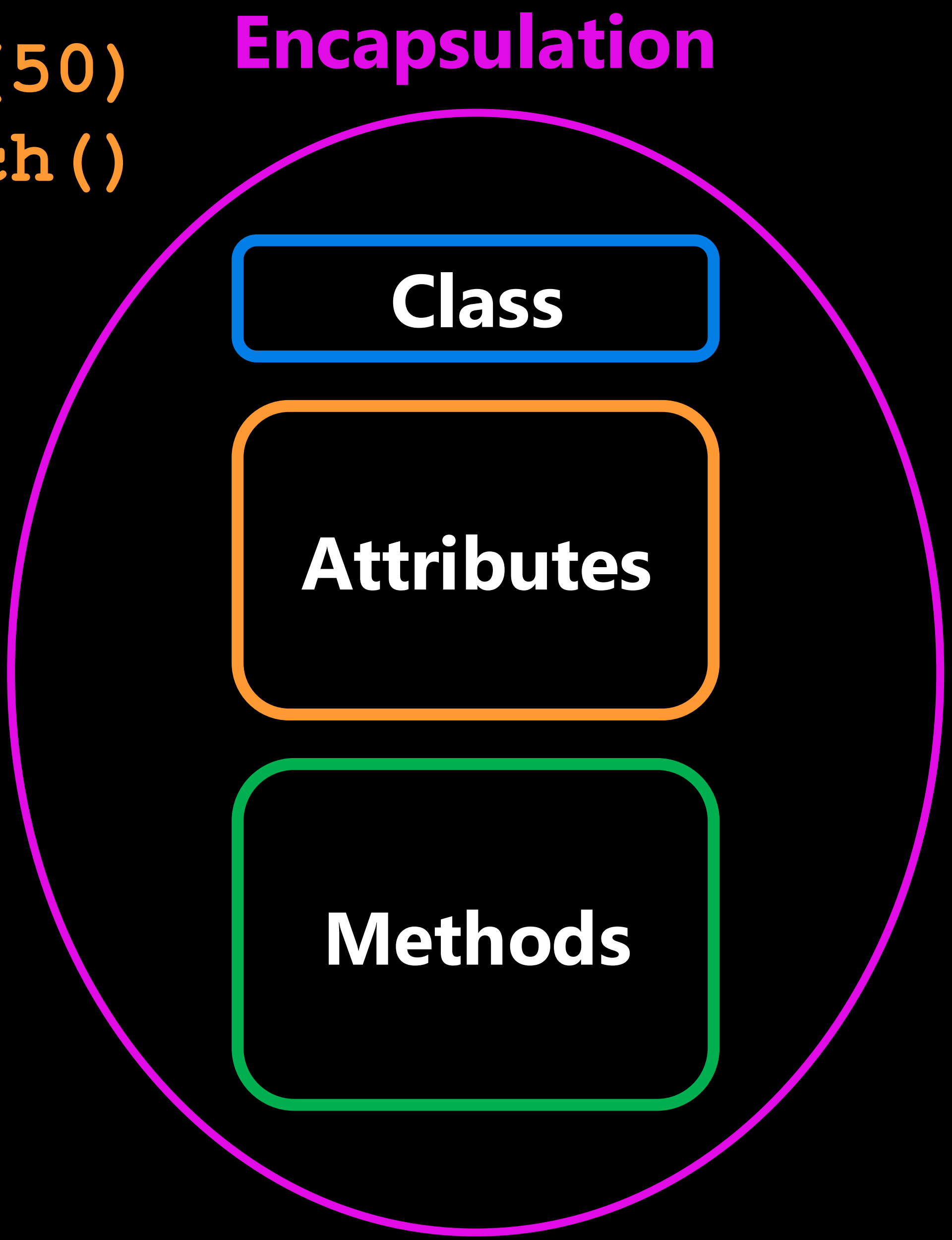
```
students = ‘Kendrick,A+\nDre:C-\nSnoop:B\n’

# create a file
myfile = open("grades.txt", "w")

# store string to the file
myfile.write(students)

# close the file
myfile.close()
```



Encapsulation

```
seb.forward(50)
kitty.scratch()
```

- The core of object-oriented programming is the organization of the program by **encapsulating** related data and functions together in an object.
- To encapsulate something means to enclose it in some kind of container.
- In programming, encapsulation means keeping **data** and the **code** that uses it in one place and hiding the details of exactly how they work together.



Encapsulate it!

- For example, each instance of class `file` keeps track of which file on the disk it is reading/writing and where it currently is on that file.
- The class hides the details of how it is done, so we (as programmers) can use it without needing to know how it is implemented

```python
f = open('test.txt','w')
f.write('hola')
f.close()
```


```python
print(f)
```

```
<_io.TextIOWrapper name='test.txt' mode='w' encoding='UTF-8'>
```

Can be any name!

Writing and Reading

```
myfile = open('test.txt', 'w')
```

'w' OR 'a'

```
myfile = open('test.txt', 'a')
```

```
myfile.write('CATS!…')
```

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Different ways of Reading a File

- Reading a file is similar to writing a file. First we need to open a file for reading (**"r"**):

```
myfile = open('test.txt', 'r')
```

- If file doesn't exist? **ERROR**
- Then to read a file we apply one of the following approaches which take advantage of various read methods:

1. The `read` approach
2. The `readline` approach
3. The `for line in file` approach
4. The `readlines` approach

No correct approach!
Multiple methods to help
with contexts and purposes

Different ways of Reading a File

Approach	Code	When to use it
The read approach	myfile = open(filename, 'r') contents = myfile.read() myfile.close()	When you want to read the whole file at once and use it as a single string.

The readline approach	myfile = open(filename, 'r')
	contents = ''
	line = myfile.readline()
	while line != '':
	    contents += line
	    line = myfile.readline()
	myfile.close()
		When you want to process only part of a file. Each time through the loop line contains one line of the file.

The for line in file approach	myfile = open(filename, 'r') contents = '' for line in myfile: contents += line myfile.close()	When you want to process every line in the file one at a time.
The readlines approach	myfile = open(filename, 'r') lines = myfile.readlines() myfile.close()	When you want to examine each line of a file by index.

TODAY

FUTURE

Let's Code!

- Let's take a look at how this works in Python!
  - Different read approaches
    - read()
    - readline()
    - for line in file
    - readlines()

Open your notebook

Click Link:
3. Reading Files

MAD LIBS™
World's Greatest Word Game

- Madlibs is a story game
- A story is written, and a few important words are taken out, replaced by blanks
- The blanks are labelled with their part of speech or other category ("noun", "adjective", "an animal", and so on)
- Replace the categories with specific words without knowing the story
- When all the blanks have been filled in, the story is read out, usually with comic results
- Example: https://youtu.be/kM9Wuzj4k24

Let's Code!

- Let's take a look at how this works in Python!
  - Bringing it all together with Madlibs!

Open your notebook

Click Link:
4. Madlibs

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introduction to Object Oriented Programming, and the file object.

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